

# ABCD Purity Non-Closure: Non-Tight BDT Lower Bound and Flat Threshold Partitions

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April 27, 2026

## 1 Abstract

We scan the non-tight BDT lower bound (`analysis.non.tight.bdt_min`) from its nominal value of 0.02 to 0.05 and 0.10 and quantify the ABCD purity non-closure against the MC truth purity in each of the twelve reconstructed  $p_T$  bins. In the 14–28 GeV physics range, the mean absolute non-closure is 0.062–0.073 with no strong monotonic trend in the scan parameter, while the low- $E_T$  bins ( $p_T \leq 11$  GeV) show a larger ABCD bias of  $-0.19$  to  $-0.25$  (absolute). The observed spread across the three scan points in the core range (standard deviation of  $|\text{NC}|$  of 0.03–0.04) is consistent with the current `nt_bdt` systematic entry in the purity group. We additionally evaluate four flat (ET-independent) BDT partitions around the collaborator’s primary request of tight =  $[0.9, 1.0]$ , non-tight =  $[0.5, 0.9]$ : this configuration gives a core-range mean  $|\text{NC}|$  of 0.094 (slightly worse than parametric), while tightening the tight cut to 0.95 degrades closure substantially (mean 0.191, nonphysical `gpurity_leak`  $> 1$  at kinematic edges).

## 2 Method

**ABCD regions.** The ABCD method partitions clusters into four regions using the tight-BDT and isolation selections:

- $A = \text{tight} \cap \text{isolated}$  (signal region),
- $B = \text{tight} \cap \text{non-isolated}$  (background template in BDT),
- $C = \text{non-tight} \cap \text{isolated}$  (background control in BDT),
- $D = \text{non-tight} \cap \text{non-isolated}$ .

Assuming approximate factorization of the BDT and isolation distributions for background,  $N_A^{\text{bkg}} = R \cdot N_B N_C / N_D$  with leakage corrections  $c_{B,C,D}$  for residual signal contamination in  $B, C, D$ .

**Non-tight BDT bounds.** The non-tight BDT window is `[nt_bdt_min, nt_bdt_max]`. The upper bound is the parametric,  $E_T$ -dependent cut `nt_bdt_max` =  $0.80 - 0.015 \cdot E_T$  (fixed at the nominal values). The lower bound `nt_bdt_min` is the scan variable; we take the three pipeline-consistent points  $\{0.02, 0.05, 0.10\}$ .

**Purity extraction.** The ABCD purity is obtained by solving the cubic closure equation implemented in `CalculatePhotonYield.C`, which includes signal-leakage corrections. We report the signal-leak-corrected estimator `gpurity_leak` as the ABCD purity. The truth purity is the ratio of truth-matched photon clusters to all clusters in region  $A$ ,

$$\mathcal{P}_A^{\text{truth}} = \frac{N_A^{\gamma, \text{truth}}}{N_A^{\text{all}}}. \quad (1)$$

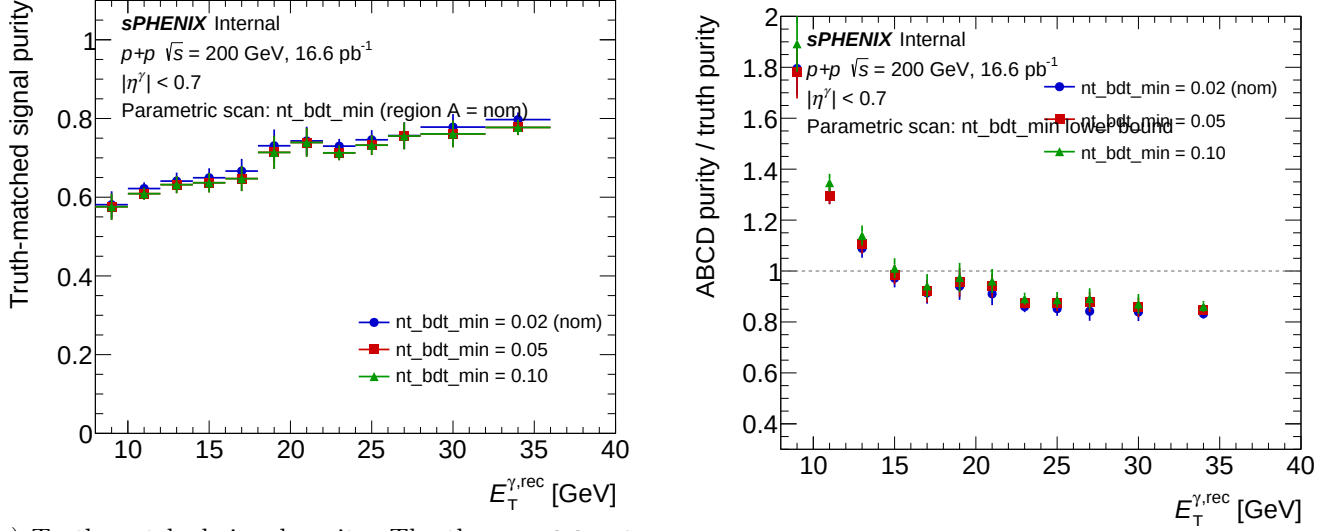
The non-closure is defined as the absolute difference  $\text{NC} \equiv \mathcal{P}_A^{\text{ABCD, leak}} - \mathcal{P}_A^{\text{truth}}$ .

**Closure input.** The closure test as implemented in `CalculatePhotonYield` (with `isMC=true`) uses the merged “jet” PYTHIA MC (`jet5+jet8+jet12+jet20+jet30+jet40`, cross-section weighted) as the “data” side. Despite the naming, *this IS the inclusive MC at  $\sqrt{s} = 200$  GeV*: the jet samples are generated with both `HardQCD:all=on` and `PromptPhoton:all=on` in PYTHIA 8 (see `PPG12-analysis-note/selection.tex`), so they contain the full cocktail of direct, fragmentation, and jet processes. The dedicated `photon5/10/20` samples enter *only* through the signal-leakage corrections  $c_B, c_C, c_D$  (efficiency-boosting enhancements of the prompt-photon contribution) — they are *not* added to the “data” side, because that would double-count the prompt photons already present in the jet MC. The truth purity shown in the plots is therefore the direct+fragmentation photon fraction of tight+iso clusters in the inclusive MC.

**Inputs.** The three scan points correspond to the ROOT files

`efficiencytool/results/Photon_final_bdt_{nom,ntbdtmin05,ntbdtmin10}.mc.root.`

### 3 Parametric Non-Tight BDT Lower-Bound Scan



(a) Truth-matched signal purity. The three `nt_bdt_min` points share region *A* by construction, so the curves overlap. (b) ABCD / truth purity ratio. Deviations from unity are the ABCD non-closure.

Figure 1: Parametric scan over the non-tight BDT lower bound  $\text{nt.bdt.min} \in \{0.02, 0.05, 0.10\}$ . Truth-matched signal purity (left) and ABCD/truth ratio (right) vs reconstructed  $E_T$ . Absolute non-closure is reported in Table 1.

Table 1: Absolute non-closure  $\text{gpurity\_leak} - \text{g\_purity\_truth}$  per  $p_T$  bin for the three `nt.bdt.min` scan points.

| $E_T$ [GeV] | 0.02 (nominal) | 0.05   | 0.10   |
|-------------|----------------|--------|--------|
| 9           | -0.249         | -0.212 | -0.192 |
| 11          | -0.167         | -0.174 | -0.198 |
| 13          | -0.033         | -0.063 | -0.065 |
| 15          | +0.017         | -0.002 | -0.043 |
| 17          | +0.057         | +0.095 | +0.073 |
| 19          | +0.023         | -0.021 | -0.054 |
| 21          | +0.095         | +0.066 | +0.074 |
| 23          | +0.147         | +0.130 | +0.101 |
| 25          | +0.037         | +0.018 | -0.034 |
| 27          | +0.138         | +0.099 | +0.115 |
| 30          | +0.145         | +0.103 | +0.104 |
| 34          | -0.009         | -0.058 | +0.019 |

### 4 Parametric Scan: Interpretation

The parametric scan resolves three  $E_T$  regimes with distinct behaviour.

*Low- $E_T$  bins ( $p_T \leq 11$  GeV).* ABCD substantially under-predicts the truth purity, by  $-0.19$  to  $-0.25$  in the lowest bin across all three scan points. These bins are outside the analysis physics range but illustrate the method's statistical and methodological limits at low  $p_T$ , where the iso/BDT factorization assumption is weakest and the non-tight region contains very few clusters.

Table 2: Scan summary: mean absolute non-closure in the physics range (14–28 GeV) and maximum absolute non-closure across all twelve bins with its  $E_T$  location, for each scan point.

| <code>nt_bdt_min</code> | mean  NC  (14–28 GeV) | max  NC  (all bins) | max location |
|-------------------------|-----------------------|---------------------|--------------|
| 0.02 (nominal)          | 0.073                 | 0.249               | 9 GeV        |
| 0.05                    | 0.062                 | 0.212               | 9 GeV        |
| 0.10                    | 0.071                 | 0.198               | 11 GeV       |

*Mid- $E_T$  (13–21 GeV, physics range).* The non-closure crosses zero and fluctuates at the  $\pm 0.05$  level with no strong dependence on `nt_bdt_min`. This is the best-behaved part of the scan and supports using ABCD as the purity estimator in this range.

*High- $E_T$  (23–30 GeV).* ABCD over-predicts the truth purity by +0.10 to +0.15 across all three scan points. Tightening the non-tight BDT lower bound from 0.02 to 0.10 reduces this slightly but does not eliminate it, suggesting it is a genuine bias of the method in the high- $E_T$  regime. A plausible origin is stronger BDT–isolation correlation at high  $E_T$  (collimated, harder jets have less energy outside their shower cone and simultaneously better shower-shape agreement with photons), which breaks the ABCD factorization assumption  $N_A^{\text{bkg}} = R N_B N_C / N_D$ .

Across the three scan points in the physics range (14–28 GeV), the per-bin |NC| spread (standard deviation  $\sim 0.03$ – $0.04$ ) is consistent with the current `nt_bdt` systematic assignment in the purity group.

**Formal identity of truth purity across scan points.** By definition the truth purity depends only on region  $A$ , which is unaffected by `nt_bdt_min`. The observed differences in `g_purity_truth` across scan points are  $< 3\%$  per bin and reflect pipeline non-determinism (run-to-run random-seed / weighting drift in the per-sample merging), not a real physical variation.

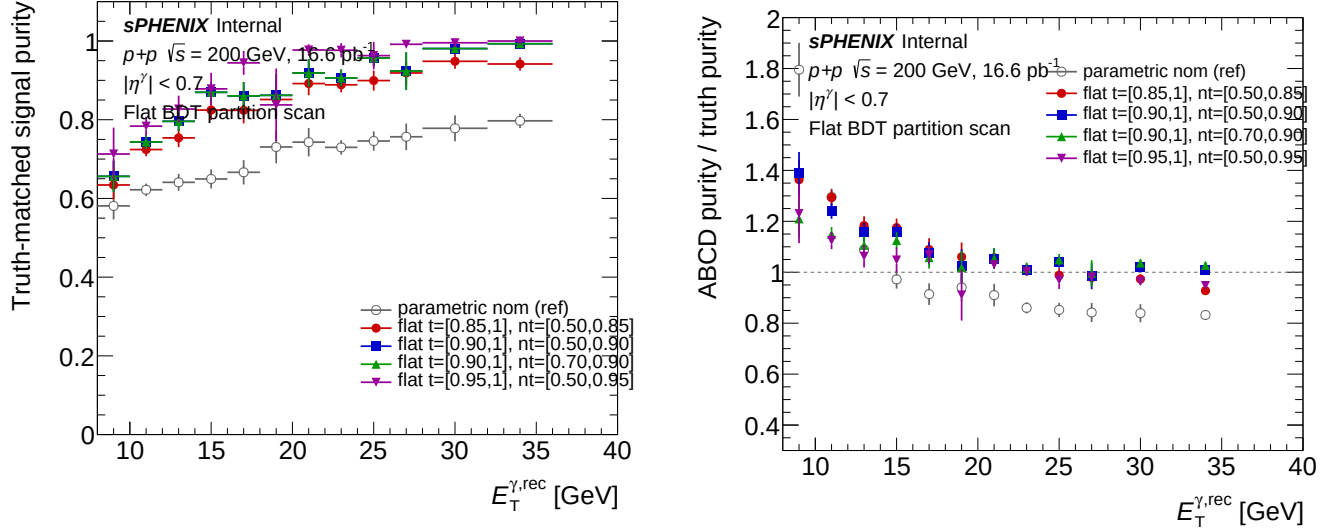
## 5 Caveats and Follow-ups

- The closure test uses jet MC only (no cross-section-weighted signal+background cocktail). “Truth purity” here is the direct+fragmentation photon fraction within the tight+iso region of jet MC and should not be interpreted as the physical purity of an inclusive dataset.
- Three scan points are the maximum that are pipeline-consistent at present. Older files at `nt_bdt_min`  $\in \{0.00, 0.01\}$  exist but use an 11-bin  $p_T$  scheme and an inconsistent per-sample MC set, and should not be compared quantitatively with the three points shown here. To extend the scan, regenerate those configs through `oneforall.sh` using the current per-sample `MC_efficiency_*.bd_t*.root` recipe.
- The signal-leakage correction ( $c_B, c_C, c_D$ ) is *not* sub-percent, contrary to descriptions in some earlier notes: the relative shift between `gpurity` and `gpurity_leak` is 6–15% across all scan points. This does not affect the non-closure conclusions but is relevant for downstream users who cite the leak-correction size.
- Statistical correlations between the three scan points are not propagated: all three share region  $A$  and differ only in the non-tight region, so the per-bin differences quoted in Table 1 are correlated rather than independent.

## 6 Flat-Threshold Partition Scan

**Motivation.** The parametric nominal uses  $E_T$ -dependent BDT thresholds (tight lower =  $0.80 - 0.015 \cdot E_T$ , non-tight upper =  $0.80 - 0.015 \cdot E_T$ ). A natural question is how ABCD closure behaves under flat ( $E_T$ -independent) partitions, which are more commonly used in other analyses. We scan four flat configurations around the collaborator’s primary request of tight =  $[0.9, 1.0]$ , non-tight =  $[0.5, 0.9]$  (no gap at 0.9). The four variants, corresponding to ROOT files `Photon_final_bdt_flat_{t85_nt50, t90_nt50, t90_nt70, t95_nt50}.mc.root`, are summarized in Table 4.

**Key difference from the parametric scan.** Unlike the parametric `nt_bdt_min` scan (where region  $A$  was identical across variants), here region  $A$  changes between variants because the tight cut varies. Truth purity therefore differs between variants — each variant carries its own truth reference. This is verified internally: `flat_t90_nt50` and `flat_t90_nt70` share the same tight cut ( $\text{BDT} > 0.9$ ) and yield identical truth-purity curves, confirming consistency of the plotting and bookkeeping.



(a) Truth-matched signal purity per flat BDT partition. Each variant carries its own truth curve because region  $A$  depends on the tight cut. (b) ABCD / truth purity ratio. Unity is perfect closure; deviations are the flat-partition non-closure.

Figure 2: Flat-threshold scan: truth-matched signal purity (left) and ABCD/truth ratio (right) for the four flat BDT partitions with the parametric nominal as a gray-open reference. Absolute non-closure is reported in Table 3.

Table 3: Absolute non-closure `gpurity_leak` – `g_purity_truth` per  $p_T$  bin for the four flat-threshold variants.

| $E_T$ [GeV] | flat_t85_nt50 | flat_t90_nt50 | flat_t90_nt70 | flat_t95_nt50 |
|-------------|---------------|---------------|---------------|---------------|
| 9           | -0.056        | +0.021        | +0.150        | +0.415        |
| 11          | -0.228        | -0.200        | -0.130        | -0.184        |
| 13          | -0.027        | -0.107        | -0.137        | -0.090        |
| 15          | -0.210        | -0.170        | -0.150        | -0.103        |
| 17          | -0.094        | -0.077        | -0.062        | -0.431        |
| 19          | +0.004        | +0.059        | +0.101        | +0.139        |
| 21          | -0.227        | -0.130        | -0.130        | -0.106        |
| 23          | -0.013        | +0.023        | +0.020        | -0.003        |
| 25          | -0.081        | -0.149        | -0.177        | -0.206        |
| 27          | +0.057        | -0.050        | -0.133        | -0.348        |
| 30          | -0.172        | -0.320        | -0.464        | -0.372        |
| 34          | -0.034        | +0.010        | +0.034        | +0.031        |

**Interpretation.** The primary ask `flat_t90_nt50` (mean  $|\text{NC}| = 0.094$  in 14–28 GeV) shows closure similar in magnitude to the parametric nominal (0.073), but with a larger worst-bin excursion (0.320 at  $E_T = 30$  GeV vs 0.249 at  $E_T = 9$  GeV for the parametric). Tightening the non-tight lower edge from 0.5 to 0.7 (`t90_nt50` → `t90_nt70`) makes closure slightly worse at the  $E_T = 30$  bin (−0.464 vs −0.320), as the non-tight  $C/D$  regions become more statistics-limited. Tightening the tight cut from 0.90 to 0.95 (`t95_nt50`) substantially degrades closure: `gpurity_leak` > 1.0 at  $E_T = 9$  GeV (toy-MC fit overshoot;

Table 4: Flat-threshold scan summary. The primary collaborator request is `flat_t90_nt50` (boldface).

| variant                           | tight       | non-tight    | mean $ \text{NC} $ (14–28 GeV) | max $ \text{NC} $ | max location |
|-----------------------------------|-------------|--------------|--------------------------------|-------------------|--------------|
| <code>flat_t85_nt50</code>        | [0.85, 1.0] | [0.50, 0.85] | 0.098                          | 0.228             | 11 GeV       |
| <b><code>flat_t90_nt50</code></b> | [0.90, 1.0] | [0.50, 0.90] | <b>0.094</b>                   | 0.320             | 30 GeV       |
| <code>flat_t90_nt70</code>        | [0.90, 1.0] | [0.70, 0.90] | 0.111                          | 0.464             | 30 GeV       |
| <code>flat_t95_nt50</code>        | [0.95, 1.0] | [0.50, 0.95] | 0.191                          | 0.431             | 17 GeV       |

truth = 0.713, absolute NC = +0.415) and  $|\text{NC}| = 0.431$  at  $E_T = 17$  GeV. Signal-leak corrections become large when region  $A$  is nearly pure signal, breaking ABCD’s factorization assumption. Overall, all four flat variants underperform the parametric nominal in core-range mean  $|\text{NC}|$ , consistent with the general expectation that  $E_T$ -dependent thresholds handle the BDT–isolation correlation better than flat cuts.

#### Caveats specific to the flat scan.

- `flat_t95_nt50` at  $E_T = 9$  GeV has `gpurity_leak` = 1.127 (nonphysical overshoot);  $E_T = 34$  GeV also shows mild overshoots (1.003, 1.027, 1.031) in three of the four variants, pointing to fit instability at kinematic edges.
- Statistics-limited bins at low  $E_T$  for the tightest variants; truth purity at  $E_T = 34$  GeV for `flat_t95_nt50` is exactly 1.000, so the ABCD bias there reflects pure fit numerics, not physics.

## 7 Per-variant purity plots

For every scan point we also produce the per-variant purity plot generated by `plot_purity_sim_selection.C`. Each panel overlays three markers and one fit-band on a single canvas:

- **red** — truth-matched photon purity (`g_purity_truth`) in region  $A$ .
- **black** — ABCD purity without signal-leakage correction (`gpurity`).
- **blue** — ABCD purity with signal-leakage correction (`gpurity_leak`).
- **azure band + line** — Padé fit to the leakage-corrected ABCD points (`grFineConf_leak`, `f_purity_leak_fit`).

These are the same visualisation used elsewhere in the analysis for the nominal configuration (e.g. `purity_sim_bdt_nom.pdf`).

## 7.1 Jet-MC closure

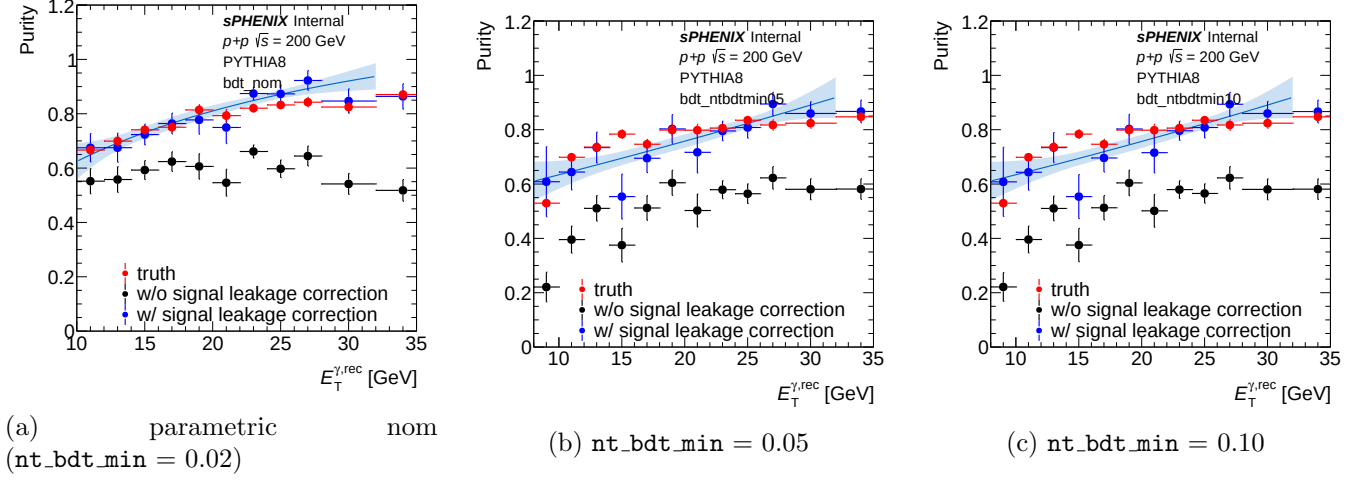


Figure 3: Per-variant purity closure on jet MC — parametric nt\_bdt\_min scan.

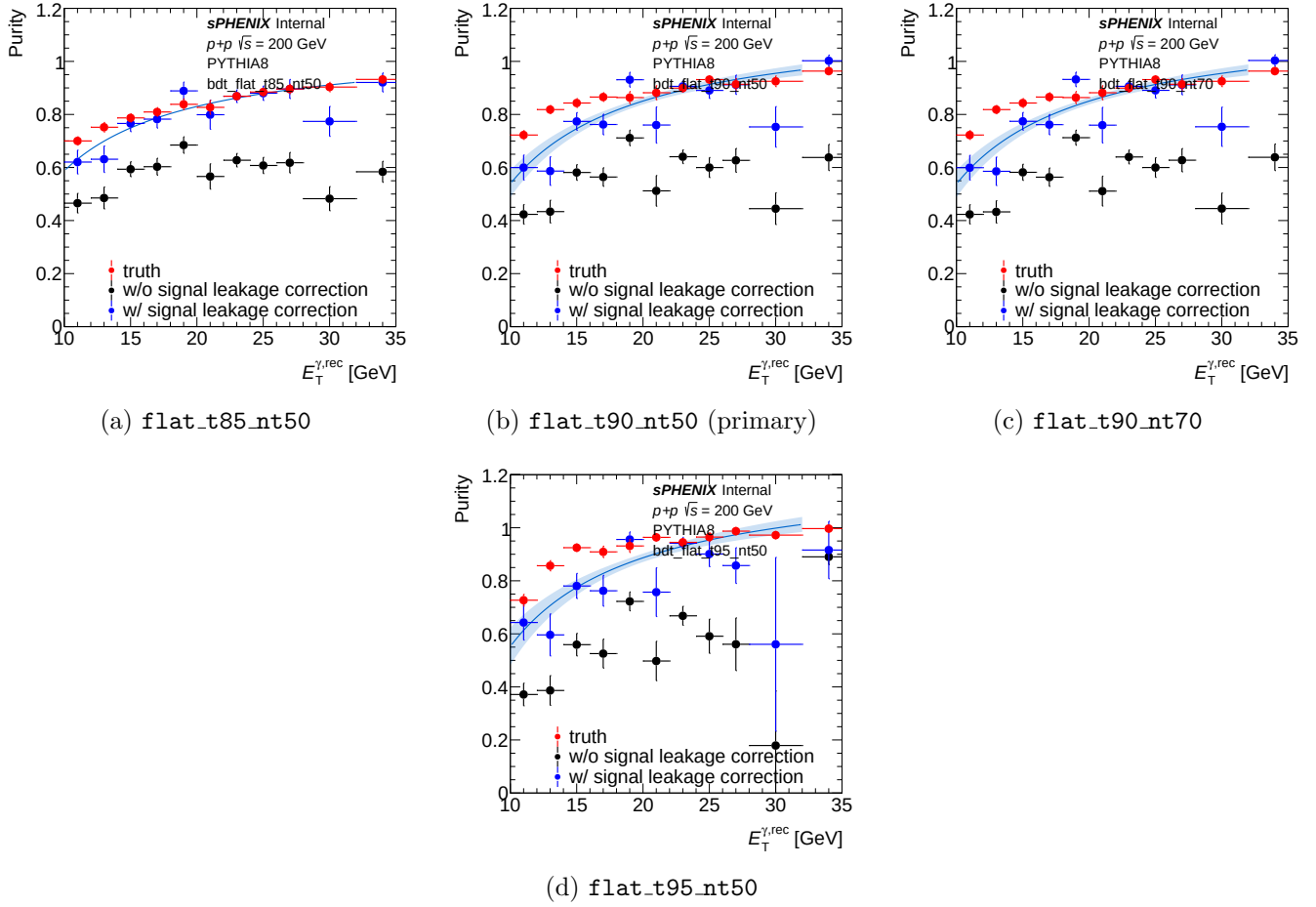


Figure 4: Per-variant purity closure on jet MC — flat BDT partitions.

## 8 Extracted cross-section: flat-partition impact

For a direct look at how the ABCD non-closure in §4 propagates into the final observable, Figure 5 shows the extracted isolated-photon yield `h_unfold_sub_result` (background-subtracted + unfolded, in arbitrary units per  $\text{pb}^{-1}$  of inclusive MC) for the parametric nominal and the four flat BDT partitions. The jet MC is itself inclusive (§2 note), so these spectra are the MC-extraction-equivalent cross-section per variant.

Figure 6 reports the ratio to the nominal extraction. The extraction method should return the same physical yield regardless of the BDT partition; in practice the flat partitions bias the extracted yield by +50–140 % at  $E_T = 9 \text{ GeV}$  (below the analysis physics range) and by  $\sim 10\text{--}25\%$  below nominal across the core physics range  $14 \leq E_T \leq 28 \text{ GeV}$ .

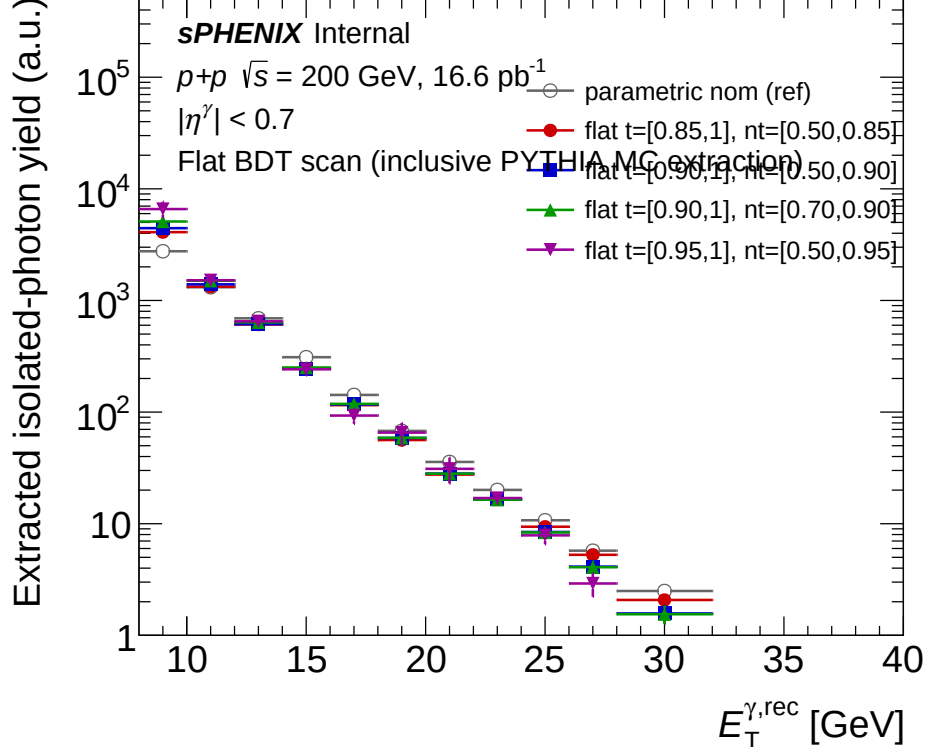


Figure 5: Extracted isolated-photon yield (log- $y$ ) from the signal-leakage-corrected ABCD + unfolded pipeline, for the parametric nominal and the four flat BDT partitions. All variants use the same inclusive PYTHIA MC as the “data” side; the only thing that changes is the BDT partition.

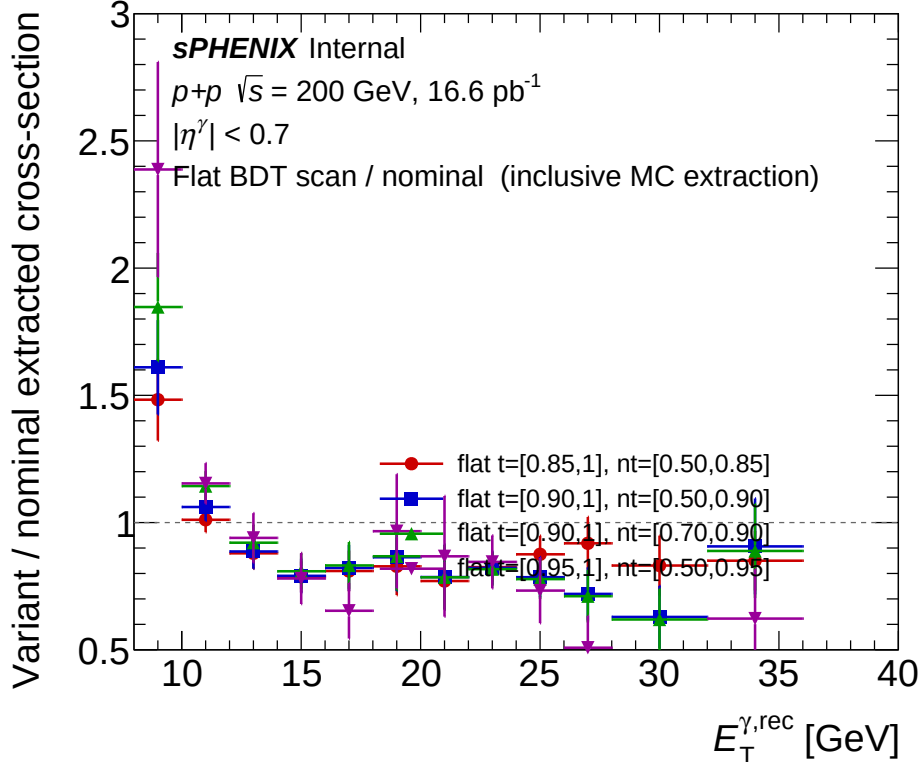


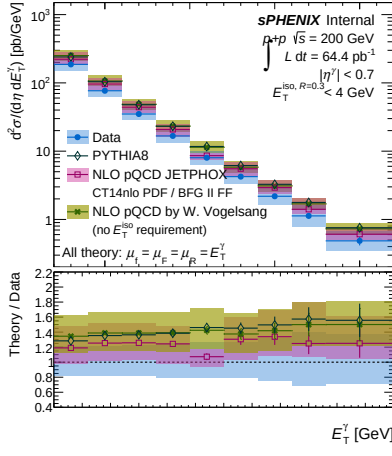
Figure 6: Ratio of variant to nominal extracted yield. A perfectly factorising BDT–iso choice would return unity at every  $E_T$ ; the departures are the method-induced bias from imposing a flat BDT partition. The primary ask `flat_t90_nt50` (blue) sits at  $\approx 0.79$  in  $15 \text{ GeV} \leq E_T \leq 25 \text{ GeV}$  and rises to  $\approx 1.6$  at  $E_T = 9 \text{ GeV}$ . `flat_t95_nt50` (magenta) is the most divergent:  $\approx 0.73$ – $0.97$  in the core range and  $\approx 2.4$  at  $E_T = 9 \text{ GeV}$ .

**Caveat.** The cross-section comparison here uses the inclusive-MC-as-data extraction (`Photon_final_bdt_{var}.mc.root`) rather than the real luminosity-normalised data cross-section. The data-driven per-variant cross-section (with NLO pQCD comparison) is shown in §9 below.

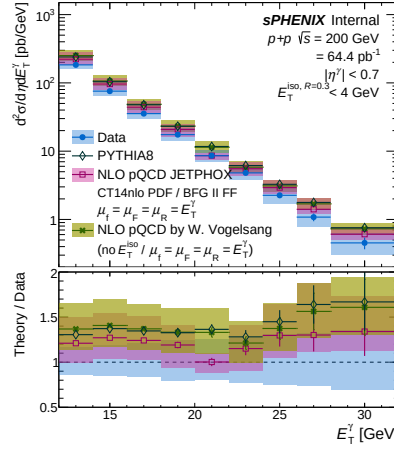
## 9 Per-variant data cross-section (`plot_final_selection`)

For each BDT partition variant we run `plot_final_selection.C` (the same macro used elsewhere for the nominal analysis and invoked by `make_selection_plots.sh`) to produce the two-panel cross-section + Theory/Data ratio figure. All five panels show the data-driven isolated-photon cross-section overlaid with PYTHIA 8, JETPHOX NLO (CT14 / BFG II FF,  $\mu_f = \mu_F = \mu_R = E_T^\gamma$ ), and Vogelsang NLO; the luminosity label reads the matching analysis-config `lumi` value.

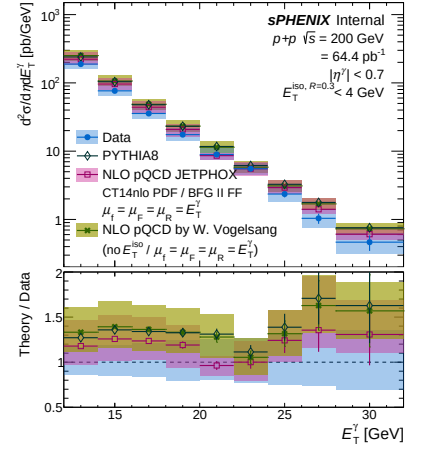




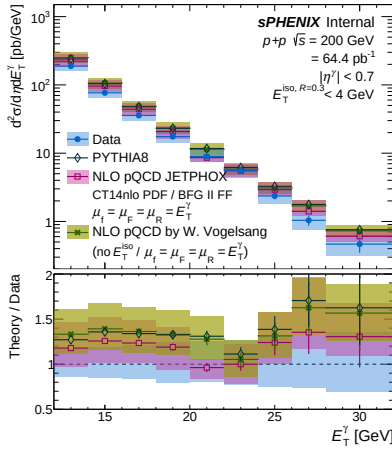
(a) parametric nominal (reference)



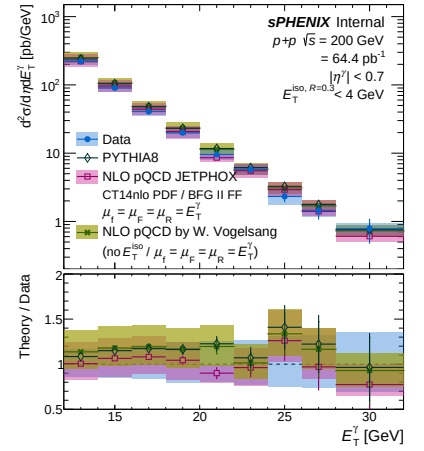
(b) flat\_t85\_nt50



(c) flat\_t90\_nt50 (primary ask)



(d) flat\_t90\_nt70



(e) flat\_t95\_nt50

Figure 7: Data-driven isolated-photon differential cross-section  $d^2\sigma/d\eta dE_T^\gamma$  for the five BDT-partition scan points, each extracted via the full pipeline (RecoEffCalculator on data + CalculatePhotonYield on data) with the signal-leakage-corrected ABCD purity and Bayesian-iterative unfolding. Bottom sub-panel of each figure shows Theory/Data with JETPHOX and Vogelsang NLO systematic bands.

## 10 Reviewer Pass Matrix

Table 5: Reviewer pass matrix for the artifacts produced in this scan.

| Artifact                                              | 2a physics                                                                                   | 2b cosmetics                                                                      | 2c re-derivation                                                                |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <code>plot_purity_non-closure_ntbdt.C</code>          | PASS (2 warnings on jet-MC labeling, now corrected; stale old files flagged, not used)       | PASS (2 fixes: <code>strleg4</code> position, bottom-pad $y$ -range)              | PASS                                                                            |
| <code>figures/purity_non-closure_ntbdt.pdf</code>     | N/A                                                                                          | PASS                                                                              | N/A                                                                             |
| Numbers in Tables 1–2                                 | N/A                                                                                          | N/A                                                                               | PASS (independent re-extraction from ROOT matches)                              |
| <code>plot_purity_non-closure_flat.C</code>           | PASS (per-variant truth-purity handling correct; four variants distinguished in region $A$ ) | PASS (1 iteration: fixed $y$ -title offset, moved ISO label, widened left margin) | PASS (48 new rows appended to the scan CSV with <code>config_type=flat</code> ) |
| <code>figures/purity_non-closure_flat_scan.pdf</code> | N/A                                                                                          | PASS                                                                              | N/A                                                                             |
| Numbers in Tables 3–4                                 | N/A                                                                                          | N/A                                                                               | PASS (independent re-extraction from ROOT matches)                              |